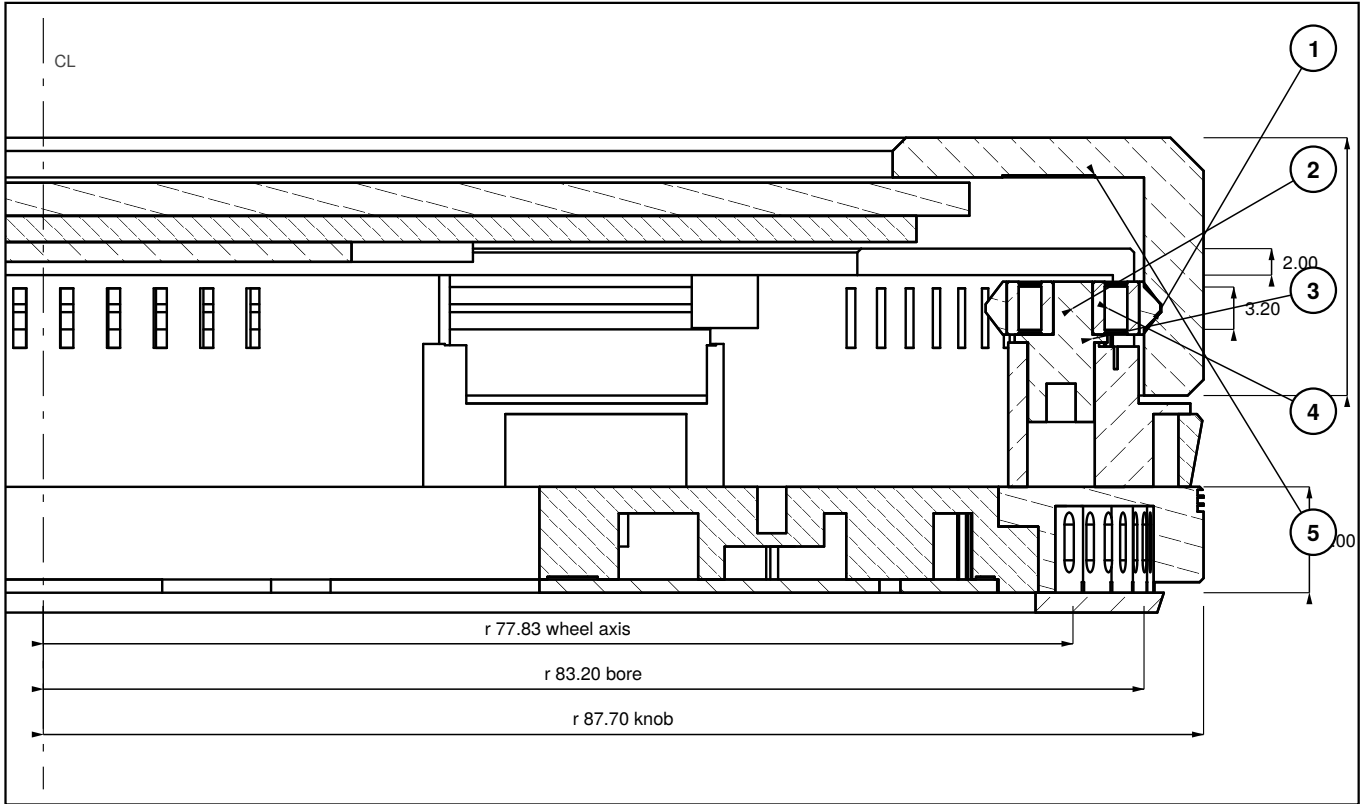
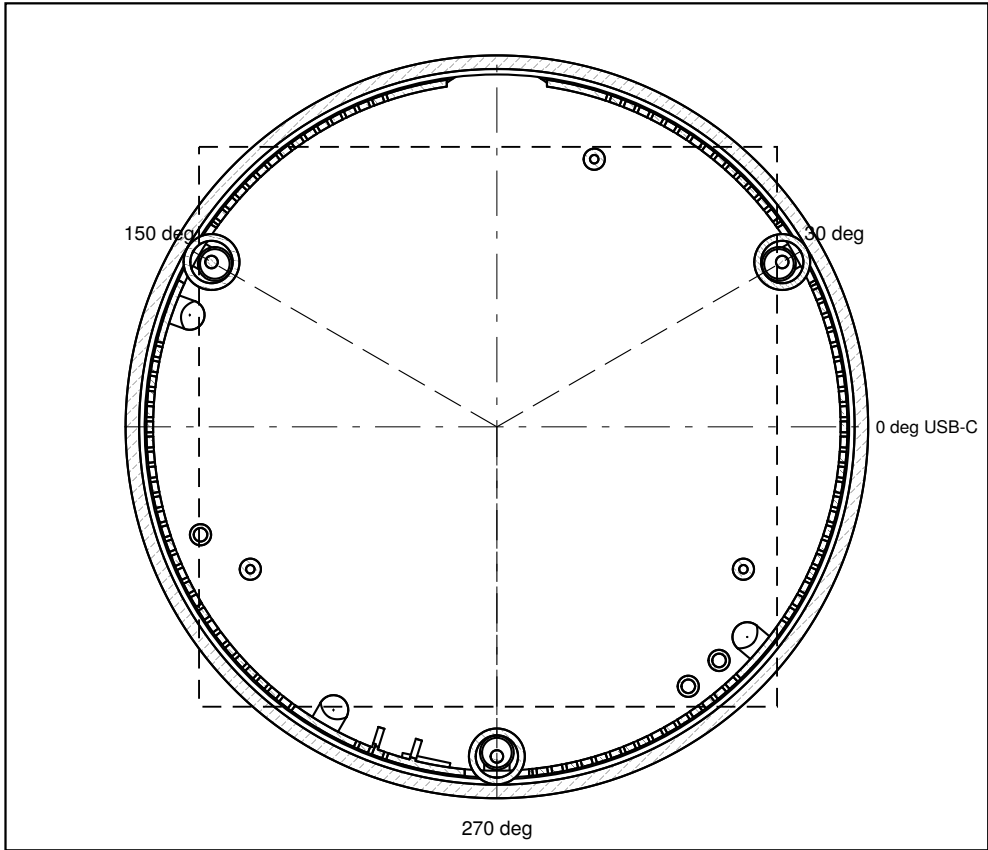


SECTION A-A scale 1.75 : 1



Right half, on the 30 deg - 210 deg axis, through the wheel at 30 deg. Hatching is cut material; plain lines are behind the cut.

PLAN - HORIZONTAL SECTION AT z 21.5 scale 0.56 : 1



Dashed rectangle: the display panel, 136.5 x 132.2, centre 2.1 in -x - its real outline from the model, above this cut.
v15 clears the panel by height, not by azimuth: the panel sits on the seat flange at z 26.0, above the posts, whose tops are at z 18.9. The posts' inner faces are at r 72.93.
The wheels stay at 30 deg / 150 deg / 270 deg so the encoder tower at 310 deg and the port slot at 0 deg have the gaps between them. Encoder on D02.

MODELLED, NOT MADE — no v15 part has been produced. Dimensions are model values, not inspection results. Bodies tagged ENVELOPE or ASSUMED in the model are drawn as envelopes, not as the real part.
Projected from docs/v15/the60_v15_assembly.step by hidden-line projection, with dimension text written from docs/v15/src/params.py. No hand drafting. The knob is drawn on its turned profile: the knurl is on the part but is not drawn.

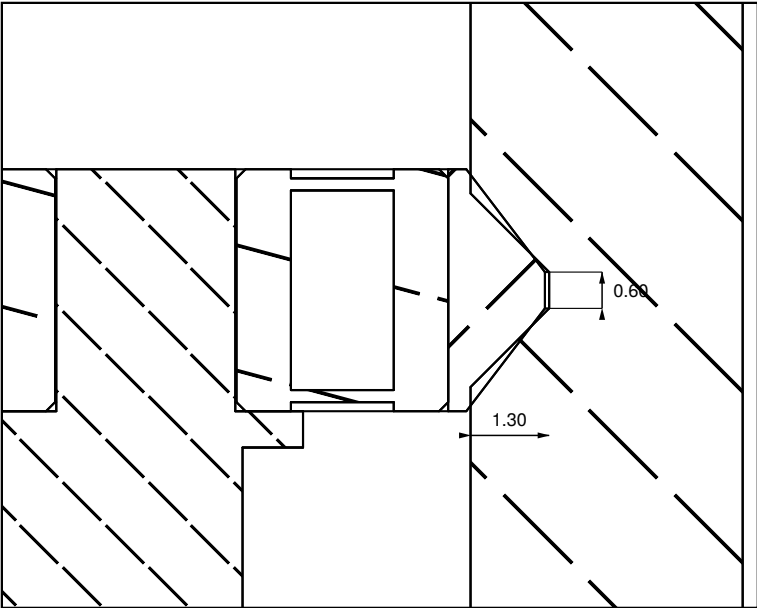
NOTES

- 1 V-groove in the bore: root r 84.50, 90 deg included, 1.30 deep, 0.60 flat at the root so contact is on the flanks only. z 19.9 - 23.1. The bore is dia 166 and smooth elsewhere: v15 moved the code ring off the bore into the crown's underside, so nothing else breaks the turned surface.
- 2 623ZZ bearing 3 x 10 x 4 pressed into a turned dia 13.20 V-collar. Axis r 77.83, z 19.5 - 23.5. Three off, at 30 deg / 150 deg / 270 deg.
- 3 Eccentric bush: a dia 5.00 shank in the post carrying a dia 2.96 wheel pin offset 0.90 from it, under a dia 7.00 flange 0.60 thick, with a 2 mm hex socket underneath. Pin inward, the wheel sits at r 76.03 - 0.57 clear inside the bore - and the knob drops on; half a turn from below seats it and sets the preload.
- 4 v15 inverted the structure. There is no ledge: the seat flange (z 24.0 - 26.0) is the datum face, and the wheel posts hang below it, dia 8.00 with a dia 5.10 bush bore, top face z 18.9. The flange's top face is the display's seat and prints flat on the bed. Plate screws from below at 160 deg / 240 deg / 320 deg.
- 5 The wheels carry the knob's weight and its lift-off load. Nothing else supports it: the drive band is a friction contact on the same bore, and the halo, the code ring and the display all clear the knob.

V-on-V locates the knob radially and axially at once and takes the lift-off load, so the knob cannot be pulled off the device.

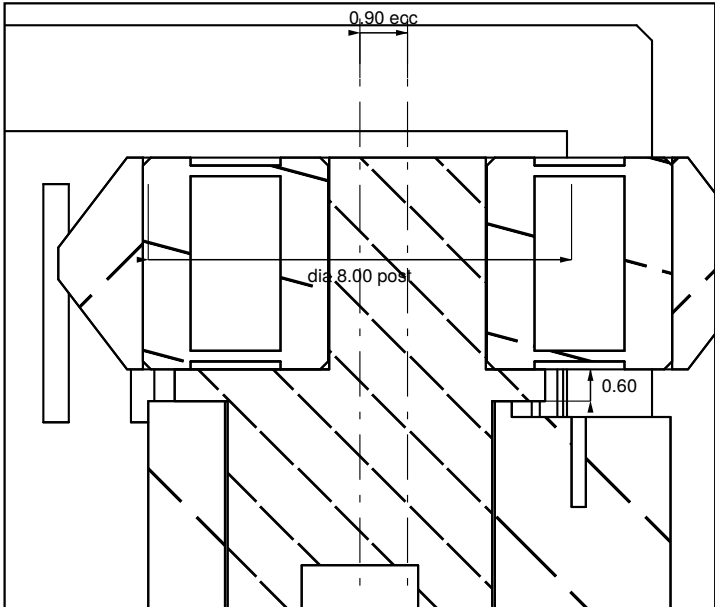
Knob 19.5 tall: 57 % of the 34.4 body height, 54 % of the 35.9 including the pad.

DETAIL X - GROOVE AND COLLAR 8 : 1



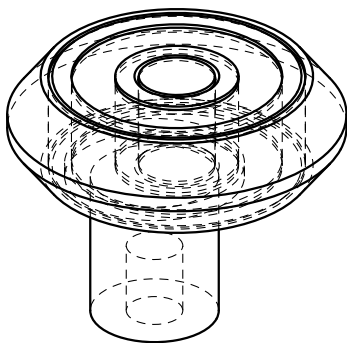
Notchiness three times a turn means the collars are bottoming on the 0.60 root flat. Bore-to-groove concentricity within 0.05 total: runout modulates the wheel preload, and the wheels are what hold the knob.

DETAIL Y - BUSH AND POST 7 : 1



Bush flange 0.60 thick sits on the post's underside at z 18.9.
Printed pin is the known weak point - brass with an M2 grub screw in production.

WHEEL AND BUSH - ISOMETRIC 3.4 : 1



V-collar over the 623ZZ, on the eccentric bush, as assembled. Hidden edges dashed.

KNOB SUPPORT - WHEELS, COLLARS AND ECCENTRIC BUSHES

the 60 (Cadrane) — desk dial

DRAWING
60-15-D01

REV
v15

SCALE
AS SHOWN

DATE
2026-09-07

UNITS
mm